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Protective film

Abstract

A protective film comprising a release liner, a pressure sensitive adhesive layer, and a base material film laminated in this order in a thickness direction thereof, wherein the base material film contains polyurethane or poly(vinyl chloride), and information is printed on an opposite side, of the release liner, from the pressure sensitive adhesive layer.

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Field of Classification Search

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) Priority is claimed on Japanese Patent Application No. 2022-052063, filed Mar. 28, 2022, the content of which is incorporated herein by reference.

(2) The present invention relates to a protective film.

Description of Related Art

(3) In recent years, personal digital assistants, such as smartphones and tablets, have become prevalent rapidly. Many of these personal digital assistants have display screens of touchscreen-type liquid crystal displays. Thus, these personal digital assistants employ a structure, in which a display screen occupies almost entirety of the front face of a body of the device. However, such a liquid crystal display is fragile against impact and is susceptible to a break or damage due to a drop or external impact. Therefore, protective films to protect display panels of personal digital assistants have been widely used.

(4) Patent Document 1 describes a protective film having a PET layer under a reinforced glass film, an adhesive layer for adhering the reinforced glass film and the PET layer, and a tacky layer for attaching the PET layer and a surface of the portable electronic device.

CITATION LIST

Patent Literature

(5) Patent Document 1: JP 2013-242567 A

SUMMARY OF THE INVENTION

(6) In recent years, for example, a smartphone having a curved display, in which the display screen has a curved shape in its end portions, have appeared. However, in a case where a film lacking flexibility, such as the reinforced glass film in Patent Document 1, is used, the protective film does not adequately conform to a display screen.

(7) Meanwhile, for the purpose of product management of a protective film and the like, a tag or label indicating a lot number or a readable code such as a bar code may be attached to the protective film.

(8) However, if the multiple protective films are transported or stored in a form of a stack during distribution and storage, the protective film may develop a mark (hereinafter, referred to as “pressed mark”) originated from a difference in heights of tag or label parts. In a case where a protective film has a base material film having high flexibility, the film is susceptible to such a pressed mark.

(9) The present invention was completed to solve the problems describe above, and an object of the present invention is to provide a protective film having high conformability to an adherend surface and preventing occurrence of pressed marks.

(10) The present invention has the following aspects.

(11) (1) A protective film including a release liner, a pressure sensitive adhesive layer, and a base material film laminated in this order in a thickness direction thereof, in which the base material film contains polyurethane or poly(vinyl chloride), and information is printed on an opposite side, of the release liner, from the pressure sensitive adhesive layer.

(12) (2) The protective film according to (1) above, further including a surface coating layer on an opposite side, of the base material film, from the pressure sensitive adhesive layer.

- (13) (3) The protective film according to (1) or (2) above, where the release liner is whitening-treated.
- (14) (4) A protective film including a release liner, a pressure sensitive adhesive layer, a base material film, and a cap film laminated in this order in a thickness direction thereof, the base material film containing polyurethane or poly(vinyl chloride), in which the base material film contains polyurethane or poly(vinyl chloride), the cap film is releasable from the protective film after the release liner has been released and removed from the protective film and the protective film is attached to an adherend, and information is printed on an opposite side, of the cap film, from the base material film.
- (15) (5) The protective film according to (4) above, further including a surface coating layer on an opposite side, of the base material film, from the pressure sensitive adhesive layer, in which the release liner, the pressure sensitive adhesive layer, the base material film, the surface coating layer, and the cap film are laminated in this order in the thickness direction thereof.
- (16) (6) The protective film according to (4) or (5) above, in which the release liner is whitening-treated.
- (17) (7) The protective film according to any one of (4) to (6) above, further including a primer layer on an opposite side, of the cap film, from the base material film.
- (18) (8) The protective film according to any one of (1) to (7) above, in which a total thickness, in which the release liner is not included, is from 110 to 600 μm .
- (19) (9) The protective film according to any one of (1) to (8) above, where the release liner is released and removed, and the protective film is adhered to a display screen of a device.
- (20) According to the present invention, a protective film having high conformability to an adherend surface and preventing occurrence of pressed marks can be provided.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross-sectional view illustrating an example of a protective film according to an embodiment of the present invention.
- (2) FIG. 2 is a cross-sectional view illustrating an example of a protective film according to an embodiment of the present invention.
- (3) FIG. 3 is a cross-sectional view illustrating an example of a protective film according to an embodiment of the present invention.
- (4) FIG. 4 is a cross-sectional view illustrating an example of a protective film according to an embodiment of the present invention.
- (5) FIG. 5 is a cross-sectional view illustrating an example of a protective film according to an embodiment of the present invention.
- (6) FIG. 6 is a cross-sectional view illustrating an example of a protective film according to an embodiment of the present invention.
- (7) FIG. 7 is a cross-sectional view illustrating an example of a protective film known in the art.
- (8) FIG. 8A is a schematic view explaining a method for producing a display body according to an embodiment of the present invention.
- (9) FIG. 8B is a schematic view explaining a method for producing a display body according to an embodiment of the present invention.
- (10) FIG. 9A is a schematic view explaining a method for producing a display body according to an embodiment of the present invention.
- (11) FIG. 9B is a schematic view explaining a method for producing a display body according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(12) Hereinafter, a protective film according to an embodiment of the present invention will be described.

(13) The protective film according to an embodiment of the present invention can protect a surface of an adherend by being attached to the adherend. The protective film according to the embodiment undergoes release and removal of the release liner in the end and attached to an adherend. In the present specification, a “protective film” may also refer to a protective film in a form where a release liner has been removed.

Protective Film

First Embodiment

(14) The protective film of the present embodiment includes a release liner, a pressure sensitive adhesive layer, and a base material film laminated in this order in a thickness direction thereof, in which the base material film contains polyurethane or poly(vinyl chloride), and information is printed on an opposite side, of the release liner, from the pressure sensitive adhesive layer.

(15) FIG. 1 is a cross-sectional view illustrating an example of a protective film according to the present embodiment. As illustrated in FIG. 1, the protective film **1a** of the present embodiment includes a release liner **10**, a pressure sensitive adhesive layer **20**, and a base material film **30** laminated in this order in a thickness direction thereof, and information is printed on an opposite side, of the release liner **10**, from the pressure sensitive adhesive layer **20**. Note that the printed information is illustrated as a printed layer **70**.

(16) Note that “on an opposite side, of the release liner **10**, from the pressure sensitive adhesive layer **20**” means that the information is printed at a position on a side that is opposite to a side of the pressure sensitive adhesive layer **20** with respect to the release liner **10**, and printing is not required to be in a direct contact with the release liner **10**.

(17) FIG. 7 is a cross-sectional view illustrating an example of a protective film **2** known in the art. On the protective film **2**, a label **L** to which information is printed as a printed layer **70** is attached to the release liner **10**. When the multiple protective films are transported or stored during distribution and storage in a form of a stack, the protective film may develop a pressed mark caused by a step due to a thickness of the label **L**.

(18) However, in the protective film of the present embodiment, because information is printed on a face that is on a side opposite to a side of the pressure sensitive adhesive layer of the release liner, the formation of the pressed mark is prevented in the protective film.

(19) Each layer constituting the protective film of the present embodiment will be described in detail below.

(20) Release Liner

(21) In the present embodiment, the release liner **10** is in direct contact with and laminated to the surface of the pressure sensitive adhesive layer **20**, and the release liner **10** is releasable from the pressure sensitive adhesive layer **20**. On the surface of the release liner **10**, the surface being in contact with the pressure sensitive adhesive layer **20**, a release agent such as a silicone may be coated. The release strength required to release the release liner **10** is not particularly limited but is preferably a release strength of a degree that an operator can peel the release liner **10** lightly with hands. The protective film **1a** is attached to an adherend in a state where the release liner **10** has been released and the pressure sensitive adhesive layer **20** is exposed.

(22) The constituent material of the release liner **10** is not particularly limited. The release liner **10** preferably contains a resin as a constituent material. Examples of the resin include polyesters such as poly(ethylene terephthalate) (PET); and polyolefins such as polypropylene and polyethylene.

(23) The release liner **10** preferably contains poly(ethylene terephthalate). The content proportion of the poly(ethylene terephthalate) with respect to 100% mass of the release liner **10** is preferably 50 mass % or greater, and more preferably 80 mass % or greater. The release liner **10** is more preferably a poly(ethylene terephthalate) film containing only poly(ethylene terephthalate) as the resin.

(24) The release liner is preferably a release liner that has undergone whitening treatment (coloring treatment into white). Examples of the whitening treatment include, but not limited to, blending of a pigment, cavity formation, and sandblast.

(25) The color of the release liner after coloring is preferably a color that facilitates easy reading of the printed information. As the release liner, a white release liner that has undergone whitening treatment is preferred. With the release liner colored in white, printing in deep color can be employed for good visibility. In a case where the release liner has been subjected to whitening-treatment, the color of the printing is preferably a color other than white and is more preferably gray or black.

(26) Examples of the white pigment in a case where the release liner is colored in white include titanium oxide, zinc oxide, precipitated barium sulfate, calcium carbonate, silica, precipitated silica (white carbon), and plaster. One type of these may be used alone, or a combination of two or more types of these may be used. In a case where the release liner contains a resin as a constituent material, for example, by kneading a pigment into the resin, a whitened release liner can be obtained.

(27) Examples of the resin film having cavities include a film having the white pigments blended therein and being drawn to form cavities around the white pigments; where the film may be a poly(ethylene terephthalate) film, a polypropylene film, a polyethylene film, and a polystyrene film.

(28) The white release liner may have a whiteness index stipulated in JIS Z8715-1999 of 20 or greater, 50 or greater, or 55 or greater.

(29) The “whiteness index” is an indicator of whiteness and stipulated by JIS Z8715-1999, and a greater value of whiteness index indicates a greater degree of whiteness.

(30) As the method of measuring the whiteness index, the whiteness index can be measured by a reflection method in accordance with JIS Z8715-1999 by allowing incidence of a pencil of rays from the surface side of the release liner using a colorimeter (e.g., SE6000, available from Nippon Denshoku Industries Co., Ltd.).

(31) The thickness of the release liner **10** is preferably, for example, 30 μm or greater and 150 μm or less but not limited to this.

(32) Printed Layer

(33) The protective film **1a** of the embodiment includes a printed layer **70** having information printed on an opposite side, of the release liner **10**, from the pressure sensitive adhesive layer **20**. The printed layer **70** indicates the information.

(34) The information printed may be identification information of the protective film, and examples thereof include information such as numbers, including lot numbers and serial numbers; and codes such as bar codes and QR codes (trade name). The information is preferably machine-readable information that can be read by a machine.

(35) Specific form of the printing is not particularly limited, and any form known in the art can be appropriately referred. Examples of the printing method include a thermal transfer method, an inkjet method, and an electrophotography (electrostatic) method.

(36) The printed layer may be a part containing a component such as an ink that is typically used in the printing methods described above and that shows the information. The component such as ink may be deposited on or impregnated into the applied part. The printed layer is typically formed in a portion of an area with respect to the area of the surface of the release liner (or the cap film described below).

(37) The thickness of the printed layer may be, for example, 5 μm or less, or 3 μm or less. In a case where the component such as ink is impregnated into the applied part, the thickness of the printed layer can be considered as 0.

(38) Primer Layer

(39) Like the protective film **1b** illustrated in FIG. 2, the protective film of the present embodiment

may further include a primer layer **60** on an opposite side, of the release liner **10**, from the pressure sensitive adhesive layer **20**.

(40) The primer layer **60** is a layer that enables enhancement of adhesion of the printing for the application target in a case where the constituent component such as ink constituting the printing is applied with the primer layer being provided between the ink and the application target, compared to a case where the constituent component such as ink constituting the printing is applied without the primer layer. The primer layer may be also referred to as an adherable surface.

(41) In an example illustrated in FIG. 2, the primer layer **60** is in direct contact with and laminated to the surface of the release liner **10**, and the printed layer **70** is in direct contact with and laminated to the surface of the primer layer **60**. When the primer layer **60** is provided on the release liner **10**, adhesion of the printed layer **70** to the release liner **10** is enhanced.

(42) The constituent component of the primer layer **60** may be appropriately selected based on the constituent component of the ink constituting the print. For example, the constituent component may contain a resin such as an acrylic resin, a styrene resin, a polyester resin, a urethane resin, a polyester urethane resin, and an acrylic urethane resin.

(43) The thickness of the primer layer **60** may be, for example, 0.1 μm or less, or from 0.01 to 0.1 μm .

(44) Pressure Sensitive Adhesive Layer

(45) The pressure sensitive adhesive layer **20** has a function as a pressure sensitive adhesive. The protective film can be adhered to an adherend by subjecting the surface of the pressure sensitive adhesive layer **20** to pressure sensitive adhesion to the adherend. In the present embodiment, the pressure sensitive adhesive layer **20** is in direct contact with and laminated to the surfaces of the release liner **10** and the base material film **30**.

(46) The material used for the formation of the pressure sensitive adhesive layer **20** is not particularly limited. For example, the pressure sensitive adhesive layer **20** may be formed by a pressure sensitive adhesive composition containing an acrylic pressure sensitive adhesive, a rubber pressure sensitive adhesive, a silicone pressure sensitive adhesive, polyurethane pressure sensitive adhesive, or the like. One type of these pressure sensitive adhesives may be used alone, or a combination of two or more types of these pressure sensitive adhesives may be used.

(47) Examples of the acrylic pressure sensitive adhesive include those formed by cross-linking a copolymer of a (meth)acrylate monomer and a cross-linkable functional group-containing monomer using a cross-linking agent.

(48) Examples of the (meth)acrylate monomer include butyl acrylate and 2-ethylhexyl acrylate.

(49) Examples of the cross-linkable functional group-containing monomer include acrylic acid and 2-hydroxyethyl acrylate.

(50) Examples of the cross-linking agent include an isocyanate cross-linking agent, an epoxy cross-linking agent, and a metal chelate cross-linking agent.

(51) Examples of the rubber pressure sensitive adhesive include block copolymers such as styrene-isoprene-styrene and polymethyl methacrylate-polybutyl acrylate-polymethyl methacrylate.

(52) Examples of the silicone pressure sensitive adhesive include those formed by addition-reacting a polydimethylsiloxane having a vinyl group and a polydimethylsiloxane having a hydrosilyl group.

(53) Examples of the urethane pressure sensitive adhesive include those formed by cross-linking a polyurethane polyol using an isocyanate cross-linking agent.

(54) The pressure sensitive adhesive composition may contain a solvent, a plasticizer, a pressure sensitive adhesion-imparting material, a stabilizer, and the like, as necessary, besides the pressure sensitive adhesive.

(55) The thickness of the pressure sensitive adhesive layer **20** may be, for example, 10 μm or greater and 100 μm or less but not limited to this.

(56) Base Material Film

(57) In the present embodiment, the base material film **30** is in direct contact with and laminated to the surface of the pressure sensitive adhesive layer **20**. The base material film **30** contains polyurethane or poly(vinyl chloride). The content proportion of the polyurethane or poly(vinyl chloride) with respect to 100% mass of the base material film **30** is preferably 50 mass % or greater, and more preferably 80 mass % or greater. The base material film **30** is more preferably a polyurethane film or poly(vinyl chloride) film which contains polyurethane or poly(vinyl chloride) as a resin.

(58) The base material film containing polyurethane or poly(vinyl chloride) as a constituent material has good conformability to the surface of an adherend and can be suitably attached to a surface of an adherend including a curved face.

(59) Examples of the polyurethane include polyester polyurethane, polyether polyurethane, and polycarbonate polyurethane.

(60) The polyester polyurethane is formed by a reaction of polyester polyol and polyisocyanate. The polyether polyurethane is formed by a reaction of polyether polyol and polyisocyanate. The polycarbonate polyurethane is formed by a reaction of polycarbonate polyol and polyisocyanate.

(61) The poly(vinyl chloride) is a polymer having a repeating unit derived from vinyl chloride. The poly(vinyl chloride) may be a homopolymer of vinyl chloride or may be a copolymer of vinyl chloride and a monomer that is copolymerizable with the vinyl chloride. Examples of the copolymer include copolymers mainly containing vinyl chloride, such as an ethylene-vinyl chloride copolymer, a vinyl acetate-vinyl chloride copolymer, and a vinyl chloride-halogenated olefin copolymer.

(62) The modulus of elasticity in tension at 23° C. of the base material film **30** may be, for example, from 10 to 1500 MPa, or from 10 to 1000 MPa, from the perspective of more suitably exhibiting conformability to a surface of an adherend.

(63) The modulus of elasticity in tension of the base material film **30** can be determined by performing a tensile test in which a test piece having a width of 15 mm and a length of greater than 100 mm made from a base material film **30** is stretched at a tensile speed of 200 mm/min and a distance of 100 mm between grips, to determine a modulus of elasticity in tension of the test piece in an elastic deformation region at 23° C.

(64) The thickness of the base material film **30** may be, for example, 100 μm or greater and 300 μm or less, or 130 μm or greater and 200 μm or less. When the thickness of the base material film **30** is not greater than the upper limit value described above, flexibility can be enhanced. When the thickness is not less than the lower limit value described above, scratch resistance can be enhanced.

(65) Surface Coating Layer

(66) Like the protective film **1c** illustrated in FIG. 3, the protective film of the present embodiment may further include a surface coating layer **40** on an opposite side, of the base material film **30**, from the pressure sensitive adhesive layer **20**.

(67) In the example illustrated in FIG. 3, the surface coating layer **40** is in direct contact with and laminated to the surface of the base material film **30**. The material used for formation of the surface coating layer **40** is not particularly limited but preferably contains a fluororesin or urethane resin, from the perspectives of achieving antifouling properties and self-repair properties. The fluororesin or urethane may be a cross-linked body. The content proportion of the fluororesin or urethane resin with respect to 100% mass of the surface coating layer **40** is preferably 50 mass % or greater, and more preferably 80 mass % or greater.

(68) Examples of the fluororesin include those formed by cross-linking a copolymer of fluoro olefin and hydroxyalkyl vinyl ether by an isocyanate cross-linking agent.

(69) Examples of the urethane resin include those formed by cross-linking polyester polyurethane by an isocyanate cross-linking agent.

(70) The thickness of the surface coating layer **40** is not particularly limited and, for example, is from 0.5 to 50 μm.

- (71) According to the protective film of the embodiment including a surface coating layer **40** as an outermost layer, a flexible base material film can be protected. Furthermore, the protective film of the embodiment suitably exhibits antifouling properties and self-repair properties. For example, the protective film can be suitably employed for a use where the protective film is attached and used on a touchscreen-type display screen.
- (72) The printed layer, the primer layer, the release liner, the pressure sensitive adhesive layer, the base material film, and the surface coating layer described above may be each formed from a single layer or multiple layers. In a case of multiple layers, the layers may be the same or different.
- (73) The thickness of each of the release liner, the pressure sensitive adhesive layer, the base material film, and the surface coating can be determined as an average value obtained by measuring thicknesses at randomly chosen 10 points. In a case of multiple layers, the thickness is the total thickness of the multiple layers.
- (74) Examples of a preferable combination of constituent materials of the protective film of the present embodiment include a combination of: a release liner containing a whitening-treated poly(ethylene terephthalate), a pressure sensitive adhesive layer formed by an acrylic pressure sensitive adhesive, a base material film containing polyurethane, and a surface coating layer containing a fluororesin.
- (75) Note that the protective film of the first embodiment described above may further include a cap film described below (also referred to as “cap sheet”) (it is noted, in the cap film, information may not be printed on an opposite side from the base material film).
- (76) An example of a protective film of one embodiment is a protective film including a release liner, a pressure sensitive adhesive layer, a base material film, and a cap film laminated in this order in the thickness direction thereof, in which the base material film contains polyurethane or poly(vinyl chloride), the cap film is releasable from the protective film after the release liner has been released and removed from the protective film and the protective film is attached to an adherend, and information is printed on an opposite side, of the release liner, from the pressure sensitive adhesive layer.
- (77) An example of a protective film of another embodiment is a protective film including a release liner, a pressure sensitive adhesive layer, a base material film, a surface coating layer, and a cap film laminated in this order in the thickness direction thereof, in which the base material film contains polyurethane or poly(vinyl chloride), the cap film is releasable from the protective film after the release liner has been released and removed from the protective film and the protective film is attached to an adherend, and information is printed on an opposite side, of the release liner, from the pressure sensitive adhesive layer.
- (78) The protective film of the embodiment is suitable for a use where the release liner is released and removed, and the protective film is attached to a display screen of a device and used.
- (79) Examples of the device include personal digital assistants such as smartphones, cellular phones, smartwatches, notebook computers, and tablets. Examples of the display screen include surfaces of various display modules such as liquid crystal displays (LCD) and organic light emitting diode (OLED). Preferably, the display screen is operable as a touchscreen.
- (80) The display screen may be a curved display of a shape that includes a curved surface partially or in its entirety, and may be a curved display of a shape that includes curved surfaces in its end portion.
- (81) With the exception of the release liner, the pressure sensitive adhesive layer, the base material film, and the surface coating layer of the protective film of the embodiment are preferably transparent. Transparency required may only need to enable a user to visually confirm contents of a display screen of a device, when the user uses the device with the protective film being attached to the display screen of the device.
- (82) The protective film of the embodiment is preferably cut into a shape corresponding to a design feature of the device. Examples of the design feature of the device include not only a shape and

size of the screen, but also design features corresponding to parts such as a camera lens, a microphone, a speaker, and a manual operation button that are provided on or around the display screen. The protective film may be cut into a shape, with which the protective film can be attached to the display screen without covering the parts such as a camera lens, a microphone, a speaker, and a manual operation button.

Second Embodiment

(83) The protective film of the present embodiment is a protective film including a release liner, a pressure sensitive adhesive layer, a base material film, and a cap film laminated in this order in a thickness direction thereof, in which the base material film contains polyurethane or poly(vinyl chloride), the cap film is releasable from the protective film after the release liner has been released and removed from the protective film and the protective film is attached to an adherend, and information is printed on an opposite side, of the cap film, from the base material film.

(84) FIG. 4 is a cross-sectional view illustrating an example of a protective film according to the present embodiment. As illustrated in FIG. 4, the protective film **1d** of the present embodiment includes a release liner **10**, a pressure sensitive adhesive layer **20**, a base material film **30**, and a cap film **50** laminated in this order in a thickness direction thereof. The cap film **50** is releasable from the protective film **1d** after the release liner **10** has been released and removed from the protective film **1d** and the protective film **1d** is attached to an adherend, and information is printed on an opposite side, of the cap film **50**, from the base material film **30**. Note that the printed information is illustrated as a printed layer **70**.

(85) The protective film of the present embodiment is a protective film described in the first embodiment including the cap film **50** and information is printed on a face of the cap film **50**.

(86) The detailed description for the part having the same constitution as the protective film of the first embodiment described above will be omitted.

(87) In the protective film of the present embodiment, information is printed on an opposite side, of the cap film, from the base material film, and because typically the printing is thin or has no thickness, development of pressed marks on the protective film can be prevented. Even if the printing has a thickness, by having the cap film, the protective film is prevented from developing pressed marks on the flexible base material film or pressure sensitive adhesive layer.

(88) Note that “on an opposite side, of the cap film **50**, from the base material film **30**” means that the information is printed at a position on a side that is opposite to a side of the base material film **30** with respect to the cap film **50**, and printing is not required to be in a direct contact with the cap film **50**.

(89) Cap Film

(90) In the present embodiment, the cap film **50** is in direct contact with and laminated to the surface of the surface coating layer **40**. The cap film **50** is laminated in a manner that the cap film **50** is releasable from the surface coating layer **40**. The cap film **50** may be laminated utilizing wettability of the surface coating layer. Alternatively, the cap film **50** may be laminated with a releasable pressure sensitive adhesive layer provided between the cap film **50** and the surface coating layer **40**.

(91) The cap film preferably contains poly(ethylene terephthalate) or polypropylene as a constituent material and more preferably contains poly(ethylene terephthalate). The cap film **50** is preferably a biaxially stretched film and more preferably a biaxially stretched poly(ethylene terephthalate) film or polypropylene film.

(92) The biaxial stretching improves the modulus of elasticity in tension of the cap film, and can easily impart greater resistance against deformation to the cap film than that of a base material film containing polyurethane or poly(vinyl chloride).

(93) Even for a protective film of an embodiment including the base material film containing a highly flexible polyurethane or poly(vinyl chloride), a cap film **50** containing a poly(ethylene terephthalate) or polypropylene film as a constituent material, or a cap film **50** preferably

containing a biaxially stretched poly(ethylene terephthalate) film or polypropylene film, can further effectively prevent elongation of the protective film during attaching operation (such as squeegeeing and re-attaching operation for positioning) of the protective film.

(94) The modulus of elasticity in tension at 23° C. of the cap film **50** may be, for example, from 2000 to 7000 MPa, or 3000 to 6000 MPa, from the perspective of more effectively suppressing elongation of the protective film.

(95) The modulus of elasticity in tension of the cap film **50** can be determined by performing a tensile test in which a test piece having a width of 15 mm and a length of greater than 100 mm made from a cap film **50** is stretched at a tensile speed of 200 mm/min and a distance of 100 mm between grips, to determine a modulus of elasticity in tension of the test piece in an elastic deformation region at 23° C.

(96) As an example, the modulus of elasticity in tension at 23° C. of the cap film **50** is preferably greater than the modulus of elasticity in tension at 23° C. of the base material film **30**.

(97) The thickness of the cap film **50** is, for example, preferably 10 μm or greater and 150 μm or less, and more preferably 20 μm or greater and 80 μm or less, but not limited to this.

(98) When the thickness of the cap film **50** is not less than the lower limit value, effect of suppressing elongation of the protective film is more effectively exhibited. When the thickness of the cap film **50** is not greater than the upper limit value, even better conformability is achieved.

(99) The release liner **10** and the cap film **50** are each preferably a poly(ethylene terephthalate) film, and similar compositions can be exemplified for the release liner **10** and the cap film **50**. However, because it is preferable for the cap film **50** to be attached to an adherend to have more enhanced conformability than the release liner **10** does, the thickness of the cap film **50** is preferably thinner than the thickness of the release liner **10**.

(100) The cap film may be formed from a single layer or multiple layers. In a case of multiple layers, the layers may be the same or different.

(101) The thickness of the cap film can be determined as an average value obtained by measuring thicknesses at randomly chosen 10 points. In a case of multiple layers, the thickness is the total thickness of the multiple layers.

(102) In a case where the protective film includes both the release liner and the cap film, the release liner is preferably a whitening-treated release liner. Even in a case where only the release liner has undergone whitening treatment and information is printed on the cap film, readability can be enhanced.

(103) With the exception of the release liner, the pressure sensitive adhesive layer, the base material film, the surface coating layer, and the cap film of the protective film of the embodiment are preferably transparent.

(104) Note that, like the protective film **1e** illustrated in FIG. 5, the protective film of the present embodiment preferably further include a surface coating layer **40** on an opposite side, of the base material film **30**, from the pressure sensitive adhesive layer **20**. The protective film **1e** includes a release liner **10**, a pressure sensitive adhesive layer **20**, a base material film **30**, a surface coating layer **40**, and a cap film **50** laminated in this order in a thickness direction thereof.

(105) Examples of the surface coating layer include those exemplified for the surface coating layer described above.

(106) Note that, like the protective film if illustrated in FIG. 6, the protective film of the present embodiment may further include a primer layer **60** on an opposite side, of the cap film **50**, from the base material film **30**. In this case, information can be printed on an opposite side, of the primer layer **60**, from the cap film **50**. Note that the printed information is illustrated as a printed layer **70**.

(107) When the primer layer **60** is provided on the cap film, adhesion of the printed layer **70** to the cap film **50** can be enhanced.

(108) Examples of the primer layer include those exemplified for the first embodiment described above.

(109) Examples of a preferable combination of constituent materials of the protective film of the present embodiment include a combination of: a release liner containing a whitening-treated poly(ethylene terephthalate), a pressure sensitive adhesive layer being formed by an acrylic pressure sensitive adhesive, a base material film containing polyurethane, a surface coating layer containing a fluororesin, and a cap film containing poly(ethylene terephthalate).

(110) The total thickness, excluding the release liner, of the protective film of the embodiment may be, for example, from 110 to 600 μm , or may be from 160 to 430 μm .

(111) When the total thickness, excluding the release liner, of the protective film is not less than the lower limit value, impact resistance is more effectively exhibited. When the total thickness of the protective film is not greater than the upper limit value, conformability to an adherend is more effectively exhibited.

(112) Note that, in the embodiment described above, a case where information is printed on the release liner or cap film of the protective film has been illustrated; however, in a case where the protective film includes the release liner and the cap film, information may be printed on both of the release liner and the cap film.

(113) According to the protective film of the embodiments described above, when the base material film contains polyurethane or poly(vinyl chloride), the protective film can achieve good conformability to a display screen (adherend). Even when the end part of the display screen is curved, attaching can be performed suitably.

(114) However, in a case where a protective film includes highly flexible base material film having good conformability to a display, if a tag or label on which information is printed is attached, the protective film tends to be affected by the height difference, and may develop pressed marks.

(115) However, in the protective film of the embodiment, because information is printed on the release liner and/or the cap film, development of pressed marks on the protective film are prevented. Furthermore, the release liner and/or the cap film to which information is printed can be released before attachment or after attachment to an adherend, and thus the information can be used without leaving the information on the protective film at the time of use.

(116) Furthermore, according to the protective film of the embodiment including the cap film, the protective film is less likely to be elongated unintentionally during attachment, and for example, misalignment between the shape of the protective film after attachment and the shape of the design feature of the device caused by elongation of the protective film is prevented.

(117) Furthermore, according to the protective film of the embodiment including a cap film, when information is printed on the cap film, print adherence of the printing is enhanced compared to a case where information is printed on a release liner to which a release agent has been transferred.

(118) Furthermore, when the primer layer **60** is provided on the release liner and/or the cap film, print adherence of the printing to the release liner and/or the cap film is further improved.

(119) Display Body

(120) The display body of an embodiment includes a display body including a pressure sensitive adhesive layer and a base material film of a protective film described in the embodiments above on a display screen of a device. The display body of the embodiment can further include a surface coating layer and/or a cap film.

(121) Examples for each constitution of the protective film include the same constitutions for the first embodiment and the second embodiment of the protective film described above.

(122) Examples of the device and the display screen thereof include the same constitutions for the embodiments of the protective film described above.

(123) The display body of the embodiment can be produced by attaching the protective film described in the embodiments above onto the display screen of the device. The produced display body of the embodiment has high conformability to a curved surface in an end portion of the display screen of the device, prevents occurrence of pressed marks of the protective film, and is in high quality. The method for producing the display body will be described below.

(124) Method for Producing Display Body

(125) The method for producing the display body of the embodiment includes attaching, the attaching including releasing and removing the release liner from the protective film of the embodiment described above and attaching the protective film, from which the release liner has been released and removed, onto the display screen of the device.

(126) Furthermore, in a case where the protective film of the embodiment includes a cap film, the attaching described above may be performed in a state the protective film includes the cap film; and then after the attaching, releasing the cap film from the protective film attached to the adherend may be further performed.

(127) In the attaching, the protective film includes the cap film, possibility of occurrence of elongation of the base material film due to the attaching operation such as squeegeeing in the attaching can be further reduced, and attachment of the protective film in a misaligned position due to misalignment of the notch corresponding to the position of a camera lens, a microphone, a speaker, or the like is prevented. In the releasing, because the cap film can be released after the attachment of the protective film to the adherend, good conformability to the display screen (adherend) is effectively exhibited.

(128) Referring to FIGS. **8A** and **8B**, the method for producing a display body of an embodiment will be described below using a case where the protective film **1a** of the embodiment illustrated above is used as an example.

(129) As an example, first, a release liner **10** is released and removed from a protective film **1a**. Thus, information printed on the release liner (printed layer **70**) is also released and removed together with the release liner **10** (FIG. **8A**).

(130) Then, an exposed pressure sensitive adhesive layer **20** is attached to a surface of a display screen D of a device (FIG. **8B**). In this way, a display body **100a** of an embodiment including the pressure sensitive adhesive layer, the base material film, and the surface coating layer of the protective film described in the embodiment above on the display screen can be produced.

(131) Furthermore, referring to FIGS. **9A** and **9B**, the method for producing a display body of an embodiment will be described using a case where the protective film **1d** is used as an example.

(132) As an example, first, a release liner **10** is released and removed from a protective film **1d** (FIG. **9A**). Then, an exposed pressure sensitive adhesive layer **20** is attached to a surface of a display screen D of a device.

(133) After the protective film **1d** is attached to an adherend, the cap film **50** laminated on the base material film **30** is released and removed (FIG. **9B**). Thus, the information printed on the cap film **50** (printed layer **70**) is also released and removed together with the cap film **50**. In this way, a display body **100d** of an embodiment including the pressure sensitive adhesive layer and the base material film of the protective film described in the embodiment above on the display screen can be produced.

(134) Note that, also in a case where a protective film of another embodiment that is not described in the method for producing a display body is used, by performing attaching in the same manner as described above, a display body including at least a pressure sensitive adhesive layer and a base material film of a protective film, on a display screen can be produced.

(135) Method for Producing Protective Film

(136) The protective film according to an embodiment can be produced by sequentially laminating each of the layers described above so as to have a corresponding positional relationship.

(137) According to the method for producing a protective film of the embodiment, the protective films of the embodiments described above can be produced.

(138) For the method for producing a protective film, the method for producing a protective film of an embodiment will be described below.

(139) For example, first, a surface coating layer **40** is formed on a base material film **30**. The surface coating layer **40** can be formed by, for example, coating a composition containing a

fluororesin and a cross-linking agent or a urethane resin and a cross-linking agent on a surface of the base material film **30** and curing the coating.

(140) The composition may further contain a solvent. Specific examples of the solvent include ester-based solvents such as ethyl acetate and butyl acetate; ketone-based solvents such as methyl ethyl ketone and methyl isobutyl ketone; and aromatic solvents such as toluene and xylene. One type of the solvents may be used alone, or a combination of two or more types of the solvents may be used.

(141) The coating method of the composition is not particularly limited and, for example, coating can be performed by using a known coating machine such as roll coaters, knife coaters, air-knife coaters, bar coaters, blade coaters, slot-die coaters, lip coaters, and gravure coaters.

(142) After the coating, the solvent is removed from the composition by drying treatment, and thus a coating is formed. The drying conditions at this time are not particularly limited, and from the perspective of solvent removal, from 60 to 150° C. is preferred. Furthermore, the drying time may be appropriately set to a time by which drying is completed and, for example, may be approximately from 30 seconds to 3 minutes.

(143) After formation of the coating, the cap film **50** including the primer layer **60** is provided thereon. Before completion of the curing of the coating, in a state where the coating and the cap film **50** thereon are arranged, for example, by further allowing the coating to stand still for approximately 1 week, curing of the coating is completed. Alternatively, after the coating is cured, a cap film **50** may be arranged by providing a releasable pressure sensitive adhesive layer (not illustrated) therebetween.

(144) The primer layer **60** can be formed by, for example, coating a composition containing a constituent component of the primer layer onto a side for which the primer layer is formed, by a bar coating method, a gravure coating method, a knife coating method, or a die coating method.

(145) Furthermore, the pressure sensitive adhesive layer **20** is formed by coating a pressure sensitive adhesive to a release liner, and the release liner is bonded to an exposed face of the base material film **30** by providing the pressure sensitive adhesive layer **20** between the release liner and the base material film **30**, and thus, the pressure sensitive adhesive layer **20** and the release liner can be provided. Alternatively, after the pressure sensitive adhesive layer **20** is formed by coating and drying the pressure sensitive adhesive onto an exposed face of the base material film **30**, a release liner may be bonded thereto.

(146) Then, printing is performed on the primer layer of the cap film, and thus a printed layer **70** is formed. Examples of the means for forming the printed layer include those described above. In this way, the protective film of the embodiment can be obtained.

(147) Note that, like the protective films **1a** to **1c**, in a case where information is printed on a release liner, a printed layer may be formed on a release liner instead of a cap film.

EXAMPLES

(148) The present invention is described in further detail below through the presentation of examples, but the present invention is not limited by these examples. For the parts assigned with reference signs, FIG. **3** (Examples 1 and 2), FIG. **5** (Example 3), or FIG. **6** (Example 4) may be referred to.

Production of Protective Film

Example 1

(149) A solution was prepared by mixing a copolymer of fluoro olefin and hydroxyalkyl vinyl ether (ethyl acetate solution having a solid content of 20.3 mass %; hydroxyl value: 67.0 mgKOH/g resin) and an isocyanate-based cross-linking agent (solids content: 75 mass %, NCO: 13.0 mass %) to give a ratio of (NCO in the cross-linking agent)/(OH in the fluororesin) to be 1/1. This solution was coated on a base material film **30** (a polyurethane film made of polyurethane and having a thickness of 150 μm; modulus of elasticity in tension: 50 MPa) and then dried at 100° C. for 5 minutes, and thus a surface coating layer **40** having a thickness of 7 μm was formed on the base

material film **30**.

(150) An acrylic solvent-type pressure sensitive adhesive [a mixture of a main agent: 100 parts by mass of acrylate copolymer solution (solids content: 40 mass %; monomer ratio: 95 parts by mass of butyl acrylate and 5 parts by mass of acrylic acid) and a cross-linking agent: 0.5 parts by mass of isocyanate-based cross-linking agent solution (solids content: 75 mass %)] was prepared and coated onto a transparent release liner **10** (obtained by coating a silicone to a PET film having a thickness of 75 μm) by a knife coater, and thus a pressure sensitive adhesive layer **20** having a thickness of 35 μm was formed on the transparent release liner **10**.

(151) An exposed face of a base material of a laminate in which the base material film **30** and the surface coating layer **40** formed as described above were laminated was stacked on an exposed face of the pressure sensitive adhesive layer **20** of the laminate in which the release liner **10** and the pressure sensitive adhesive layer **20** were laminated and subjected to pressure bonding.

(152) On a surface of the transparent release liner **10**, bar code was printed by a thermal transfer method, and thus a protective film of Example 1 was obtained.

Example 2

(153) A protective film of Example 2 was obtained in the same manner as in Example 1 described above except for using a white release liner **10'** (PET film having a thickness of 75 μm , obtained by kneading silica particles as white pigments) in place of the transparent release liner **10** of Example 1.

(154) The whiteness index stipulated in JIS Z8715-1999 of the white release liner was 60.

Example 3

(155) The layers were laminated in the same manner as in Example 1 described above except for coating the solution used for the formation of the surface coating layer **40** in Example 2 described above onto the base material film **30** and then laminating a cap film **50** (biaxially stretched PET film having a thickness of 25 μm ; modulus of elasticity in tension: 4000 MPa) on the coating thereof.

(156) On a surface of the cap film **50**, bar code was printed by a thermal transfer method, and thus a protective film of Example 3 was obtained.

Example 4

(157) A protective film of Example 4 was obtained in the same manner as in Example 3 described above except for, in Example 3, using a cap film **50** having a primer layer **60** having a thickness of 0.06 μm , and providing a printed layer **70** by printing a code on a surface of the primer layer **60** by a thermal transfer method.

(158) The primer layer is made of a polyester resin.

Comparative Example 1

(159) A protective film of Comparative Example 1 was prepared in the same manner as in Example 3 described above except for using a transparent release liner **10** in place of the white release liner **10'** of Example 3 and attaching a white pressure sensitive adhesive label (thickness: 50 μm) on which a bar code is printed on a white base material onto a surface of the release liner, instead of printing a code on the surface of the cap film **50**.

(160) Evaluation

(161) Pressed Mark

(162) For the protective film produced in each of Examples and Comparative Example described above, 50 sheets of the protective films were stacked and stored and allowed to stand still for 7 days.

(163) Then, the protective films stacked at from the bottommost to the tenth bottommost protective films were taken out. The pressed marks due to the storage while the protective films were stacked were evaluated based on the following criteria. A: The pressed marks were not observed in all 10 sheets of the protective films F: The pressed marks were observed in some of the 10 sheets of the protective films.

Readability

(164) The bar code provided on each of the protective films produced in Examples and Comparative Example described above was read by a bar code reader (application) of a smartphone, and readability was evaluated based on the following criteria. A: There were some cases where the bar code was not recognized by the reader. B: Recognition of the bar code by the reader was good.

Print Adherence

(165) For the protective film produced in each of Examples and Comparative Example described above, a cellophane tape was attached to the printed bar code part, and the cellophane tape was removed immediately afterwards. And thus print adherence was evaluated based on the following criteria. A+: No blurring or removal of the printing was confirmed, and print adherence was excellent. A: Almost no blurring or removal of the printing was confirmed, and print adherence was good. B: Blurring or removal of the printing was confirmed in some parts.

(166) The constitutions of the protective films, and the evaluation results for the protective films described above are shown in Table 1.

(167) TABLE-US-00001

TABLE 1	Comparative Example 1	Example 2	Example 3	Example 4
Information	Printing	Printing	Printing	Printing
Label display	attachment	method	Cap	film
None	None	Present	Present	Present (Primer layer present)
Transparent (Untreated)	Release liner	Present	Present	Present
White	White	Transparent	Printed part	Release
White	White	Transparent	Release	Cap film
Pressed mark	A	A	A	A
Readability	B	A	A	A
Print adherence	B	B	A	A ^{sup.} + A

(168) In Comparative Example 1, because the pressure sensitive adhesive label on which the bar code was printed on the white base material was attached, although it had good readability of the bar code, a pressed mark having a shape corresponded to the pressure sensitive adhesive label was confirmed.

(169) Based on the shape and position of the pressed mark, it was presumed that the pressed mark was caused by the height difference of the pressure sensitive adhesive labels being pressed against the soft polyurethane film (base material film).

(170) On the other hand, for Examples 1 to 4 in which the code was printed on the release liner or the cap film, no pressed mark was confirmed.

(171) Regarding readability, Examples 2 to 4 in which the release liner was subjected to white coloring treatment exhibited improved readability of bar code compared to that of Example 1.

(172) Regarding print adherence, Examples 3 and 4 in which printing was performed on the cap film had better print adherence than Examples 1 and 2 in which printing was performed on the release liner. It is presumed that this is because printing on the cap film is less likely to be affected by the release treatment agent contained in the release liner.

(173) Furthermore, Example 4 in which the cap film included the primer layer exhibited particularly excellent print adherence.

(174) Each configuration in each of the embodiments and the combinations thereof are merely examples, and additions, omissions, substitutions, and other changes may be made to the configuration within a range that does not depart from the spirit of the present invention. The present invention is also not limited by each of the embodiments and is limited only by the scope of the claims.

Claims

1. A protective film comprising: a release liner; a pressure sensitive adhesive layer; a base material film; and a cap film, laminated in this order in a thickness direction thereof, wherein the base material film contains polyurethane or poly(vinyl chloride), the cap film is releasable from the protective film after the release liner has been released and removed from the protective film and

- the protective film is attached to an adherend, identification information is printed on an opposite side, of the cap film, from the base material film, the base material film has a modulus of elasticity in tension at 23° C. of 10 to 1500 MPa, the cap film has a modulus of elasticity in tension at 23° C. of 2000 to 7000 MPa, the release liner is whitening-treated, and the pressure sensitive adhesive layer, the base material film and the cap film are transparent.
2. The protective film according to claim 1, further comprising a surface coating layer on an opposite side, of the base material film, from the pressure sensitive adhesive layer, wherein the release liner, the pressure sensitive adhesive layer, the base material film, the surface coating layer, and the cap film are laminated in this order in the thickness direction thereof.
 3. The protective film according to claim 2, wherein the surface coating layer is in direct contact with and laminated to the surface of the base material film, and the surface coating layer contains a fluororesin or urethane resin.
 4. The protective film according to claim 3, wherein a content proportion of the fluororesin or urethane resin with respect to 100% mass of the surface coating layer is 50 mass % or greater.
 5. The protective film according to claim 4, wherein the content proportion of the fluororesin or urethane resin with respect to 100% mass of the surface coating layer is 80 mass % or greater.
 6. The protective film according to claim 3, wherein the cap film is in direct contact with and laminated to the surface of the surface coating layer.
 7. The protective film according to claim 6, wherein a content proportion of the fluororesin or urethane resin with respect to 100% mass of the surface coating layer is 50 mass % or greater.
 8. The protective film according to claim 7, wherein the content proportion of the fluororesin or urethane resin with respect to 100% mass of the surface coating layer is 80 mass % or greater.
 9. The protective film according to claim 2, further comprising a primer layer on an opposite side, of the cap film, from the base material film, wherein the surface coating layer is in direct contact with and laminated to the surface of the base material film, and the surface coating layer contains a fluororesin or urethane resin, wherein the cap film is in direct contact with and laminated to the surface of the surface coating layer, and wherein the primer layer contains a resin selected from the group consisting of an acrylic resin, a styrene resin, a polyester resin, a urethane resin, a polyester urethane resin, and an acrylic urethane resin.
 10. The protective film according to claim 1, further comprising a primer layer on an opposite side, of the cap film, from the base material film.
 11. The protective film according to claim 10, wherein the primer layer contains a resin selected from the group consisting of an acrylic resin, a styrene resin, a polyester resin, a urethane resin, a polyester urethane resin, and an acrylic urethane resin.
 12. The protective film according to claim 1, wherein a total thickness, in which a thickness of the release liner is not included, is from 110 to 600 μm .
 13. The protective film according to claim 1, wherein the release liner is released and removed, and the protective film is adhered to a display screen of a device.
 14. The protective film according to claim 1, wherein the identification information is bar codes or QR codes.
 15. The protective film according to claim 1, wherein the identification information is machine-readable identification information with printed information regarding the protective film printed directly onto the cap film without a label.
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